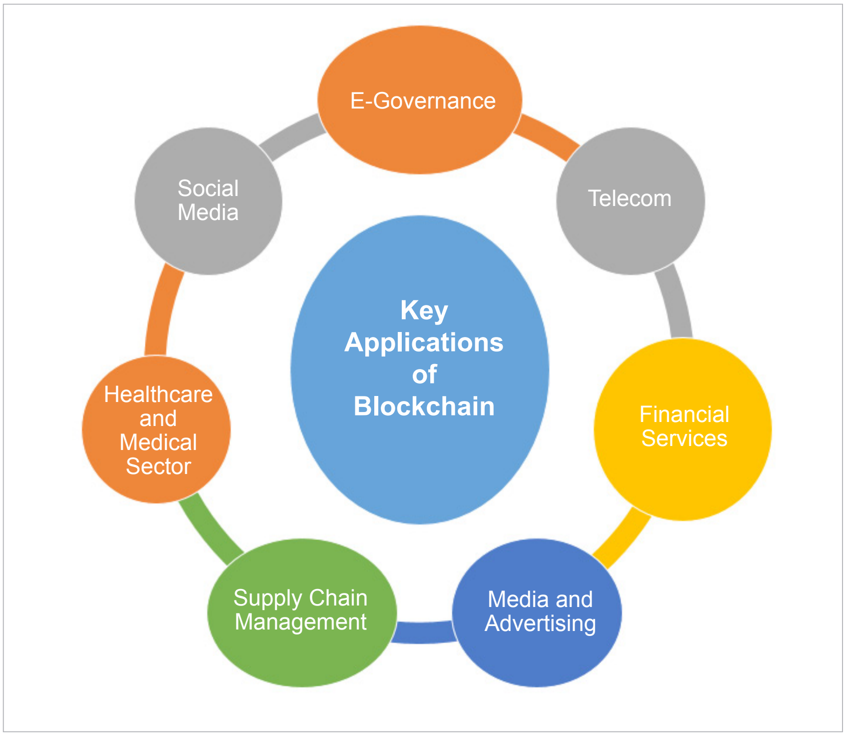


Figure 1: Applications and use cases of blockchain

Table 1: Blockchain vs traditional databases

| Blockchain | Traditional databases |
|---|---|
| Distributed ledger | Centralised database server |
| Decentralised | Single point of control and scope of failure |
| Peer-to-peer | Multiple gateways and human control |
| Cryptography based verification and authentication | Human intervention and scope of attacks |
| Resilient and fault-tolerant | Need for backups |

and connected to the next block in the chain. All subsequent data is assembled into a new block, which is added to the chain after it has been filled.

Blockchain based databases differ from traditional databases due to their decentralised nature and the underlying blockchain technology (Table 1). The data in a blockchain is stored and managed on a digital ledger, which is highly secure. There is no administrator who can exploit the records or tables in it. This is a peer-to-peer network.

The node stores the record in its database and broadcasts it if the vetting procedure by all nodes is successful. The record is then written to each node's database, ensuring that the network's status and records are always up-to-date. These features make it nearly impossible to alter or replicate the data recorded in blockchains. In order to ensure the integrity of the database, all nodes review each new entry before it is added.

The transactions in the blocks of the blockchain are hashed using cryptography, and each block establishes a connected chain with the previous one. Because no node or client can dispute a record, it provides a high degree of transparency.

## Open source database engines for blockchain development

Let's take a look at the major open source databases that can be used for development of blockchain based applications.

### BigchainDB

*(https://www.bigchaindb.com/)*
BigchainDB is a blockchain based storage system powered by MongoDB that lets you incorporate decentralised and blockchain technologies into your application. It is perhaps the most popular blockchain database because of its features including immutability and decentralisation.

The BigchainDB database is fault-proof, which implies that once a record has been confirmed and put in the database, it cannot be updated or altered in any way. With the usage of Byzantine Fault Tolerance (BFT), even if up to half of the network's nodes are defective, it can still analyse what the next block will be. For this reason, the network can erase one of the node's MongoDB databases if a hacker gets hold of it.

In addition, it can support multiple assets at once, and can hold numerous types of assets. All assets in the Bigchain network can be issued by the nodes' users. BigchainDB is a leader in supply chain management applications, where data organisation, immutability, and security are really needed.

### Apache Cassandra

*(https://cassandra.apache.org)*
This NoSQL distributed database promises linear scalability and high consistency without sacrificing speed. As a Java based distributed database system, Cassandra provides high availability without a single point of failure by spreading data over a large number of commodity computers.

Each row in Cassandra comprises tables that need a primary key, and each row is partitioned. Cassandra's partitioning allows it to split the rows